
Turn in #3

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MA 542 Section A1

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Problem 1

Let $\phi : \mathbb{Q}(i) \rightarrow \mathbb{Q}$, and suppose for contradiction that $\phi \neq 0$. Then there exists $a + bi \in \mathbb{Q}(i)$ such that $\phi(a + bi) \neq 0$. Since $a, bi \in \mathbb{Q}(i)$ and ϕ is a ring homomorphism, it follows that

$$\phi(a + bi) = \phi(a) + \phi(bi) \neq 0.$$

Note that if $1 \mapsto 0$, then for all $x \in \mathbb{Q}(i)$,

$$\phi(x) = \phi(1x) = \phi(1)\phi(x) = 0\phi(x) = 0.$$

This contradicts our assumption that $\phi(a + bi) \neq 0$, so we have that $\phi(1) \neq 0$. Suppose that $\phi(1) = r$ for some $r \in \mathbb{Q}$. Note that $4 \in \mathbb{Q}$. It follows that

$$\phi(4) = \phi(1 + 1 + 1 + 1) = 4\phi(1) = 4r.$$

Additionally,

$$\phi(4) = \phi(2)\phi(2) = (2\phi(1))(2\phi(1)) = 4r^2.$$

Therefore, $r^2 = r$, and since $r \in \mathbb{Q} \setminus \{0\}$, we have that $r = 1$. Therefore, $\phi(1) = 1$. It follows that $\phi(r) = r$ for all $r \in \mathbb{Z}$ by the same proof I gave in the previous homework, and thus $\phi(-1) = -1$.

Now consider $\phi(i)$. Note that $i^2 = -1$, so

$$-1 = \phi(-1) = \phi(i^2) = \phi(i)^2 \implies \phi(i) = \pm\sqrt{-1} = \pm i.$$

However, $\pm i \notin \mathbb{Q}$, which is a contradiction. Since $\phi(-1) = -1$ is guaranteed since $\phi(1) \neq 0$, this contradicts our original assumption that $\phi \neq 0$, and thus ϕ is the 0 homomorphism.

Problem 2

First, the commutative case. Let $\text{char } R = 2$, and let $\phi : R \rightarrow R$ be the map $x \mapsto x^2$. It follows that for all $x, y \in R$,

$$\begin{aligned} \phi(x + y) &= (x + y)^2 \\ &= x^2 + xy + xy + y^2 \\ &= x^2 + y^2 + xy(1 + 1) \\ &= \phi(x) + \phi(y) + 0xy \\ &= \phi(x) + \phi(y); \end{aligned}$$

$$\begin{aligned}
\phi(xy) &= (xy)^2 \\
&= (xy)(xy) \\
&= x^2y^2 \\
&= \phi(x)\phi(y).
\end{aligned}$$

Therefore, ϕ is a ring homomorphism.

Now suppose R is noncommutative. Suppose for contradiction that ϕ is a homomorphism. It follows that

$$\begin{aligned}
\phi(x+y)^2 &= (x+y)^2 \\
&= (x+y)(x+y) \\
&= x(x+y) + y(x+y) \\
&= x^2 + xy + yx + y^2 \\
&= x^2 + y^2 + xy + yx \\
&= \phi(x) + \phi(y) \\
&= x^2 + y^2 \\
\implies x^2 + y^2 + xy + yx &= x^2 + y^2.
\end{aligned}$$

It follows that $xy + yx = 0$, and thus $xy = -yx$ for all $x, y \in R$. Note once again that $\text{char } R = 2$. It follows that for any $r \in R$,

$$r + r = r(1 + 1) = 0 \implies r + r = 0 \implies r = -r.$$

Therefore,

$$xy = -yx \implies xy = yx.$$

Therefore, R is commutative. This contradicts the fact that R is noncommutative, and thus ϕ cannot be a homomorphism if R is noncommutative.